

Domain and Range: All Three Notations

TEACHER KEY | Algebra II Honors | Unit 1 | Day 1 | TEK A2.7.I

Objective: I can write domain and range in verbal, inequality, set, and interval notation, determine a domain algebraically from a function rule, and use union notation for a domain that comes in separate pieces.

Vocabulary

Set-builder notation — $\{x \mid \text{condition}\}$, read "the set of all x **such that** the condition holds."

Interval notation — $[]$ includes an endpoint, $()$ excludes it. Infinity always takes a **parenthesis**.

Union $\cup$ — joins two separate intervals when the domain has a **gap** in it.

Undefined — an input the rule cannot accept. Two causes in this course: a **negative radicand** and a **zero denominator**.

Part 1 — From Pairs and Tables

List least to greatest, each value once.

a. $\{(-4, -6), (0, -1), (2, 4), (3, -7), (7, 7)\}$ D **$\{-4, 0, 2, 3, 7\}$** R **$\{-7, -6, -1, 4, 7\}$**

b. $\{(-5, 3), (-1, 0), (2, 3), (6, -8)\}$ D **$\{-5, -1, 2, 6\}$** R **$\{-8, 0, 3\}$**

c. Why does the range in (b) have three values when there are four pairs? **3 repeats, and a set lists it once**

Part 2 — All Three Notations

Complete every cell.

Verbal description	Inequality	Set notation	Interval notation
all real numbers	$-\infty < x < \infty$	$\{x \mid x \in \mathbb{R}\}$	$(-\infty, \infty)$
all reals at least -6	$x \geq -6$	$\{x \mid x \in \mathbb{R}, x \geq -6\}$	$[-6, \infty)$
all reals less than 5	$x < 5$	$\{x \mid x \in \mathbb{R}, x < 5\}$	$(-\infty, 5)$
all reals from -3 to 3, 3 included only	$-3 < x \leq 3$	$\{x \mid x \in \mathbb{R}, -3 < x \leq 3\}$	$(-3, 3]$
all reals except 2	$x \neq 2$	$\{x \mid x \in \mathbb{R}, x \neq 2\}$	$(-\infty, 2) \cup (2, \infty)$
just the numbers -2, 0, and 4	(none)	$\{-2, 0, 4\}$	does not apply

The last two rows are the ones to think hardest about. Explain the difference between them:

Row 5 is continuous with one point removed; row 6 is **discrete, so intervals do not apply**

Part 3 — Finding a Domain from the Rule

No graph, no table - reason from the algebra. Ask what the rule forbids, then exclude it.

Function	What is forbidden?	Inequality	Interval notation
$f(x) = \sqrt{x - 3}$	radicand < 0	$x \geq 3$	$[3, \infty)$
$g(x) = 1/(x - 5)$	denominator $= 0$	$x \neq 5$	$(-\infty, 5) \cup (5, \infty)$
$h(x) = \sqrt{2x + 8}$	radicand < 0	$x \geq -4$	$[-4, \infty)$
$k(x) = 1/\sqrt{x - 1}$	both	$x > 1$	$(1, \infty)$

Look at $k(x)$. Why is the answer $x > 1$ and not $x \geq 1$?

At $x = 1$ the radicand is 0, and that would make the denominator **0** — so 1 must be excluded too.

Part 4 — A Domain in Three Pieces

Find the domain of $m(x) = (x + 2)/(x^2 - 9)$. Show the factoring.

Denominator factors as **$(x - 3)(x + 3)$** , so x cannot be **3 or -3**

Domain, interval notation: **$(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$**

Domain, set notation: **$\{x \mid x \in \mathbb{R}, x \neq \pm 3\}$**

Which notation communicates this domain more efficiently, and why?

Part 5 — Going Further

1. Explain why ∞ can never take a bracket, no matter the function.

2. Write a function whose domain is exactly $[4, \infty)$. **$f(x) = \sqrt{x - 4}$**

3. Write a function whose domain is exactly $(-\infty, 0) \cup (0, \infty)$. **$f(x) = 1/x$**

4. Can a domain be empty? Give a rule with no valid inputs, or argue that none exists.

Sample: **$f(x) = \sqrt{-1 - x^2}$ - the radicand is always negative**

Exit Ticket

Give the domain of $f(x) = \sqrt{x + 5}/(x - 2)$ in interval notation, and justify each restriction.

Domain **$[-5, 2) \cup (2, \infty)$**

Two reasons: **radicand needs $x \geq -5$, and the denominator forbids $x = 2$**